

## 15.1 Exam Review

Four modules. Name the method, then the number that would have caught the failure.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

The exam is cumulative. It asks you to **name a method** and **say what you would measure**. Trivia without a metric is not the point of DATA 443/643.

**First time the exam format appears.** Every first-time method this semester had a what / why / architecture / how / formula / tradeoffs block. The exam uses that.

**What.** Four modules. You name the method, then the number that would have caught the failure.

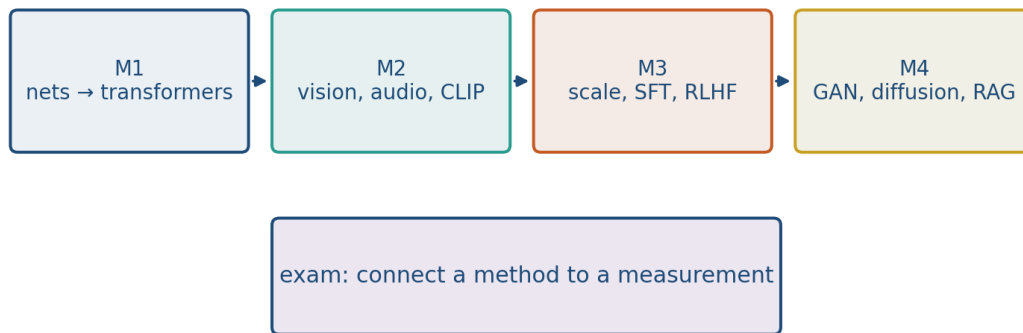
**Why.** Fluency is not a check. RAG without recall@k, GANs without coverage, CoT without faithfulness are incomplete. **Architecture.** Closed-book written exam plus remaining talks. Formula sheet is what you can write, not trivia. **How.** Practice: “same face, low loss<sub>G</sub>” → mode bins. “Cite [d5], fact in [d1]” → attribution + recall@k. **Formula.** Write the method’s equation, then the measurement (ASR, recall@k,  $D^* = 1 / 2$ ,  $\sigma(0) = 1 / 2$ , ...). **Tradeoffs.** + Cumulative and fair if you studied the first-time blocks. – Naming a paper without a metric scores poorly.

### 1. Four modules

| Module | Weeks | Spine                                                                                                    |
|--------|-------|----------------------------------------------------------------------------------------------------------|
| 1      | 1–3   | Neurons, embeddings, RNNs/gates, attention, GPT vs BERT                                                  |
| 2      | 4–6   | Multimodal, ViT patches, contrastive / CLIP / BLIP, audio as tokens                                      |
| 3      | 7–10  | Scaling and efficiency, SFT and adapters, RLHF / DPO, safety and editing                                 |
| 4      | 11–14 | GANs and mode coverage, diffusion and latent text conditioning, CoT / vote / faithfulness, RAG and ReAct |

You should be able to draw the picture (two GAN players;  $q(x_t | x_0)$ ; retrieve-then-generate) and attach **one number** to it (coverage, a denoising loss, recall@k, majority-vote accuracy, a faithfulness rate).

Four modules, one stack. The project sits on top of all four.



Exam shape: name the method, then the number that would have caught it.

| failure                 | measure                |
|-------------------------|------------------------|
| same face, low lossG    | mode bins              |
| right box, last step 52 | faithful among wins    |
| cite [d5], fact in [d1] | attribution + recall@k |

## 2. How to study

Rework the labs on CPU: XOR and embeddings, long-range gates, one attention head, toy CLIP retrieval, 2-D GAN coverage, one reverse diffusion step, vote-and-flag traces, bag-of-words RAG. Redo each note's practice questions out loud. For every named method, write one sentence of the form "I would change  $X$ ; I would report  $Y$  on split  $Z$ ."

Formulas worth being able to write:

- neuron:  $a = \sigma(w^\top x + b)$
- attention:  $\text{softmax}(QK^\top / \sqrt{d})V$
- GAN:  $\min_G \max_D \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$
- diffusion jump:  $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon$
- CFG:  $\hat{\varepsilon} = \varepsilon_u + s(\varepsilon_c - \varepsilon_u)$
- cosine:  $a^\top b / (\|a\| \|b\|)$
- DPO-shaped check:  $\sigma(0) = 1 / 2$  when the two log-ratio terms match

You do not need to memorize paper years. You do need to say **what you would plot** after you write the formula.

A compact board checklist:

| Failure you see                | Method that was in play | Number that would have caught it             |
|--------------------------------|-------------------------|----------------------------------------------|
| All fakes look like one face   | GAN                     | coverage / mode bins, not <code>lossG</code> |
| Pretty pictures, missed prompt | latent diffusion + CFG  | CLIP image–text cosine and <code>s</code>    |
| Right box, wrong algebra       | CoT                     | faithfulness rate among correct boxes        |
| Vote of a popular wrong number | self-consistency        | shared-bug check; $k$ vs accuracy plot       |
| Cite [d5], fact only in [d1]   | RAG                     | attribution + recall@k                       |
| Calc returned 408, answer 428  | ReAct                   | observation used? parse log                  |

### 3. What the exam asks

Typical item: given a failure (collapsed modes, ignored chunk, lucky CoT answer, biased CLIP hit), **name the method that was in play** and **the measurement** that would have caught it. Another item: pick GPT versus BERT, CLIP versus BLIP, GAN versus diffusion, CoT versus ReAct, and justify with the task plus the metric.

Lab → exam shortcuts: Lab 11 bins = coverage; Lab 12 oracle reverse step = you know  $x_0$ ; Lab 13 last-integer vs box = faithfulness; Lab 14 cosine rank + `calc log` = RAG + ReAct. If you can redo those four on paper, Module 4 is in shape.

Project language belongs here: baseline, one justified change, ablation, failure case. That is the same spine as note **14.3**.

### Remaining talks

**Canvas is official** for who speaks when. If you still owe a presentation, bring a **failure case** (input, output, what broke), not only a demo that worked. Upload what Canvas asks. Time limits on the hub still apply.

### 4. Teaching this note

This is a **review map**, not a new method. About **40 minutes** at the board walking the four-module table and the failure→metric grid, then **~15 minutes** of Karpathy as a recap of the LLM stack (not of GANs/diffusion). Leave 20 minutes for student questions and remaining talks. Weeks 13–14 are CoT, vote/search, faithfulness, RAG, and ReAct; Weeks 11–12 are GANs and diffusion.

Board order:

- **0–12 min.** Modules 1–2: neuron, embedding, attention, CLIP cosine. One picture each.

- **12–24 min.** Modules 3–4: SFT/RLHF/DPO in one sentence each; GAN vs diffusion; CoT vs RAG vs ReAct.
- **24–40 min.** Failure table. Worked exam-style item (GAN collapse; unfaithful CoT; RAG citation).

Then play the Karpathy minutes below. **GANs and diffusion are not in that talk**; restudy notes **11.1–12.3** and the assigned clips from those weeks.

Bring one page of handwritten formulas and one page of “failure → metric” rows. That beats rereading every HTML slide the night before.

## 5. Worked example (one exam-style item)

Prompt: “Fakes are sharp. Every fake is the same face. Generator loss is low. What do you measure?”

Answer in the form the exam wants:

- **Method:** GAN (Week 11). Implicit sampler;  $D$  can be fooled by one mode.
- **Not a measurement:** `lossG` (it can fall because  $D$  collapsed).
- **Measurement:** coverage — fraction of fakes nearest each real mode (Lab 11), or precision/recall in a feature space. Report both bins, not a mean.

A second item: “CoT accuracy 90%; a calculator disagrees with half the traces.”

- **Method:** chain-of-thought; the bug is **unfaithful** traces (Week 13; Lab 13 already scored it).
- **Measurement:** among correct boxes, fraction whose parsed steps match the box. Self-consistency may still vote the lucky number.

A third item: “Top-2 cosine missed the gold chunk; the answer still cites it.”

- **Method:** RAG. Retrieval recall@2 is 0 on that item.
- **Check:** attribution against the stuffed context, not against the whole corpus. Citing an id that was never retrieved is a fail.

Keep the answers short. The exam is not a blog post.

## 6. Where students get stuck

- Memorizing paper titles and then blanking on the metric.
- Mixing CoT / self-consistency / ToT / ReAct as one name.
- Using Karpathy as a substitute for Weeks 11–12. That video does not teach  $q(x_t | x_0)$ .

## 7. Video

Watch [Andrej Karpathy, Intro to Large Language Models](#) as a **recap of the stack**. Map weeks to minutes; do not try to play the whole hour in class.

| Minutes     | What he is doing                             | Course weeks                                            |
|-------------|----------------------------------------------|---------------------------------------------------------|
| 0:20–11:22  | Inference, then training on internet text    | 1–3 (net, tokens, next-token)                           |
| 11:22–17:52 | How they work; finetune into an assistant    | 1–3 architecture; 8 SFT                                 |
| 21:05–27:43 | Comparisons, labels, RLHF; then scaling laws | 7 scaling; 9 RLHF / DPO                                 |
| 27:43–33:32 | Tool use: browser, calculator, interpreter   | 14 RAG (browser as retrieve) and ReAct                  |
| 33:32–35:00 | Multimodality                                | 4–6 CLIP / vision / audio                               |
| 35:00–38:02 | Thinking, System 1/2                         | 13 CoT / extra tokens                                   |
| 40:45–42:15 | Custom GPTs, files                           | 14 RAG as a private index                               |
| 45:43–59:23 | Security (high level)                        | 10 red-teaming <i>as evaluation</i> , not as a cookbook |

In class, play **11:22–17:52** and **27:43–38:02** (~17 min) if you only have one pass. Students who missed Week 7–9 should add **21:05–27:43**. Students who missed Week 13–14 should add **35:00–42:15**.

Same URL as notes **13.1**, **14.1**, and **14.2**. Deep Dive chapters from **13.2–13.3** are optional extra, not a replacement for the formula sheet.

Week 11–12 video recap (not Karpathy): generative models from **46:45** ([setup / train / collapse](#)); Umar Jamil diffusion [1sPXkm2NH4](#) from **7:35**; Stable Diffusion [ZBKpAp\\_6TGI](#) **0:00–45:00**. Talk craft: Peyton Jones [sT\\_-owjKlbA](#) **0:00–20:00**.

## 8. Practice

1. A GAN's fakes look sharp and every one is the same face. What do you measure besides looking?
2. A RAG answer cites [d5] but the claim only appears in [d1]. Method name, and the check.
3. CoT accuracy is high; a calculator disagrees with half the traces. What exam idea is that, and which week's lab already scored it?
4.  $x_0 = 1$ ,  $\bar{\alpha}_t = 0.64$ ,  $\varepsilon = 2$ . Write  $x_t$ . Which week is this, and is it a GAN step?
5. Three CoT answers 408, 238, 408. Majority? If all three traces compute  $17 \times 20 = 340$  and still box those numbers, what does the vote fail to catch?